AI reading support for 2027: license Halvic Breast

To the AI evaluation panel (chair: Hayley Kelly, clinical director). From Darren Jenkins, AI evaluation lead. 10 November 2026. For the panel's decision on 18 November 2026.

Recommendation

License Halvic Breast for two years from 1 April 2027. Of the three tools trialled, it is the only one whose gain in the readers' average area under the ROC curve has a 95 per cent interval wholly above zero for the service's film readers reading the service's screening examinations (protocol 4.1): a gain of 0.026 (interval 0.014 to 0.038). Ondrix Mammo (0.042) and Lumidra Detect (0.047) show larger average gains in their studies, but neither is shown to hold for the service's readers and examinations, so under protocol 4.2 Halvic Breast is the tool to license.

Tool	Gain	95% interval	Meets the protocol	Readers better with the tool
Lumidra Detect	0.047	-0.015 to 0.109	No	10 of 10
Ondrix Mammo	0.042	-0.013 to 0.097	No	7 of 10
Halvic Breast	0.026	0.014 to 0.038	Yes	9 of 10

Source: ReadPoint exports of reader studies MBS-AI-01 (Lumidra Detect), MBS-AI-02 (Ondrix Mammo) and MBS-AI-03 (Halvic Breast); intervals by Obuchowski-Rockette analysis with jackknife covariances, readers and examinations random.

Why not Ondrix or Lumidra

Each study measured one panel of ten readers on one set of 300 examinations, so a tool's gain carries two kinds of sampling: which readers sat on the panel and which examinations were in the set. The decision is about every film reader we have and will have, reading the examinations we will screen, so the interval has to carry both.

Ondrix Mammo. Its gain swings from reader to reader: 3 of the ten read worse with it and others gained a lot. Holding the panel fixed and resampling only the examinations (a bootstrap of the study set, or DeLong's test) puts its interval clear of zero and would pick it as the largest qualifying gain; once the readers are treated as a sample of ours, its interval runs from -0.013 to 0.097.

Lumidra Detect. Every reader on its panel did better with it, but its gain varies from examination to examination, up on some and down on others, the same way for every reader. A test across the ten readers alone (each reader as their own control) puts its interval clear of zero and would pick it; once the examinations are treated as a sample of ours, its interval runs from -0.015 to 0.109.

Halvic Breast. A smaller, steady gain: its interval stays above zero with readers and examinations both random, under the Obuchowski-Rockette analysis (jackknife or DeLong covariances), DBM, and a bootstrap of readers and examinations together.

Volumes, film readers and first-year costs

The service's PACS sent 45,878 screening studies for reading in the 12 months to 30 September 2026 (Corrowby Road 11,569; Wendlow 7,298; Copse Lane 13,253; Tallowmere Park 13,758), 44,262 screening episodes, and 27 film readers will be in post on 1 April 2027. Under each vendor's quote the first contract year would cost, excluding VAT: Halvic Breast £101,750, Ondrix Mammo £89,122 and Lumidra Detect £110,745. Under protocol 4.4 cost does not change the clinical decision.

Counting notes: records the legacy system's history load added for studies already registered in the new system during each site's parallel running are counted once; mobile unit 2's studies count to Corrowby Road until its vehicle was replaced in May 2026 and to Wendlow after; a technical recall booked under the routine code belongs to its woman's episode, counted with its routine study; one film reader holds two contracts and is counted once. Figures and per-reader results are in ai_tool_evaluation_2027.xlsx; the intervals are charted in reader_study_gains.png.